MIL-F-55608(EL)
15 February 1968

MILITARY SPECIFICATION
FOR
FILTER, INFRARED, INTERFERENCE TYPE, FOR AN/TVS-3

1. SCOPE

1.1 This specification covers one type of an interference infrared filter for use in the xenon searchlight, AN/TVS-3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY :

MIL-E-17555	- Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	- Identification Marking of U. S. Military Property

FSC 5855

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DRAWINGS

SC-D-599920	- Infrared Filter - 1 Outer, Blocking
SC-D-599922	- Infrared Filter - 1 Outer, Exit
SC-D-599919	- Infrared Filter - 1 Inner, Blocking
SC-D-599921	- Infrared Filter - 1 Inner, Exit

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. - The filters shall be interference type coated in accordance with drawings SC-D-599919, SC-D-599920, SC-D-599921 and SC-D-599922 and as specified herein.

3.2 First Article Model. - The supplier shall furnish two complete filters within the time frame specified in the contract or purchase order to prove, prior to starting production, that his production methods will produce filters that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the first article model shall be subject to the approval of the contracting officer. Approval of the first article model by the activity concerned shall not relieve the supplier of his obligation to furnish filters conforming to this specification.

3.3 Initial Production. - Two filters selected from the first ten (10) filters produced from production tooling shall be examined and tested as specified in 4.4 to determine conformance to the requirements of this specification. Acceptance of initial production filters shall not establish the acceptability of the remainder of filters under contract in conflict with the acceptance levels specified in 4.4. These filters shall be fully representative of filters to be furnished under the contract.

3.4 Drawings. - The drawings forming a part of this specification are engineering design drawings. The supplier is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the filter. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. In the event of conflict between the drawings and this specification, this specification shall govern.

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3.5 Materials and Components. - Materials and components shall be as specified herein and as shown on the applicable drawings. Materials and components not specified shall be selected by the contractor and shall be subject to all applicable provisions of this specification.

3.6 Spectral Transmission. - Spectral transmission of the usable and non-usable zones of the filter shall be as follows under an ambient temperature range of 70 to 80° F and equilibrium temperature of 300° F.

3.6.1 Usable zones. - Spectral transmission of the usable zone shall be between the minimum and maximum values shown on figure 1 for any wavelength from 800 to 1100 millimicrons.

3.6.2 Non-usable zones. - No spectral transmission is required of the non-usable zone, except that it shall meet requirements of 3.8.

3.7 Interference Coating Radiation Exposure. - The spectral transmission of the filter shall meet the requirements of 3.6 when tested in accordance with 4.6.4.

3.8 Visual Security. - There shall be no visible light evident through the interference coated filter beyond 1000 meters or less than 800 meters when tested in a mechanism simulating a searchlight, AN/TVS-3 operated at the 20KW power level.

3.9 Interference Coating Bond. - The interference coating bond shall be of sufficient strength to prevent the peeling of the interference coating from the clear glass.

3.10 Environmental Requirements

3.10.1 Storage. - The filter shall not be damaged by storage at any temperature from minus 80° to plus 165° F at maximum relative humidity at any temperature within that range.

3.10.2 Operation. - The filter shall not be damaged by operation on a mechanism simulating a searchlight AN/TVS-3 in any ambient temperature from minus 65° F to plus 125° F at maximum relative humidity for any temperature within that range.

3.11 Marking. - Each filter segment shall be marked on the coated surface part in accordance with MIL-STD-130. Markings shall be contained in an area not larger than 1/8 inch wide around the outer edge of the filter segment.

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diagram of Figure 1

3.12 Workmanship. - The filter shall have no cracks, chips, striations bubbles, scratches, or any other imperfection which could impair the normal function of the filter. The coatings shall be applied uniformly to provide a uniform emission of radiated infrared energy.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. - The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards. The Government reserves the right to require the contractor to prove a stated requirement where no detailed test is specified.

4.2 Classification of Inspection. - Inspection shall be classified as follows:

a. First Article Inspection (see 4.3).

b. Initial Production Inspection (see 4.4).

c. Quality Conformance Inspection (see 4.5).

d. Inspection Comparison (see 4.7).

e. Inspection of Preparation for Delivery (see 4.8).

4.3 First Article Inspection

4.3.1 Examination. - Each filter shall be examined for defects in accordance with Table I. One or more defects shall be cause for rejection.

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TABLE I EXAMINATION

		REQUIREMENT PARAGRAPH
MAJOR		
101.	Absence of any part or component.	3.4
102.	Interference coating not applied to usable zone of filter.	3.4
103.	Filter segments cracked.	3.12
MINOR		
201.	Absence or incorrectness of identification markings.	3.11
202.	Workmanship not as specified.	3.12
203.	Dimensions not as specified.	3.4

4.3.2 Tests. - Upon successful completion of the examination specified in Table I the first article models shall be subjected to the tests in Table II. Tests shall be conducted in the order listed. Failure of any test shall be cause for rejection.

TABLE II FIRST ARTICLE TEST SCHEDULE

TEST	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Pin Holing	4.6.1	3.8
Spectral Transmission	4.6.2	3.6
Environmental Requirements	4.6.3	3.10
Interference Coating Radiation Exposure	4.6.4	3.7
Visual Security	4.6.5	3.8
Interference Coating Bond	4.6.6	3.9

4.4 Initial Production Inspection. - When specified (see 3.3), two initial production filters will be selected at random by the Government from the first 10 filters being produced from production tooling and shall be examined as specified in Table I and shall be tested as specified in Table II to determine conformance to the requirements of this specification. The inspection will be performed by the contractor under the supervision of a representative of the contracting officer. Acceptance of any initial production filters shall not exclude the remaining filters from the quality conformance inspection and acceptance provisions specified herein.

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4.4.1 Inspection failure. - Failure of an initial production filter to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production filter and shall be cause for refusal by the Government to continue acceptance of production filters until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on filters previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all filters accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality Conformance Inspection.

4.5.1 Examination. - Each filter shall be examined as specified in Table I. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests

4.5.2.1 Group A. - Each filter shall be tested in accordance with Table III. Failure of any test shall be cause for rejection.

TABLE III GROUP "A" TEST SCHEDULE

TEST	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Pin Holing	4.6.1	3.8
Interference Coating Bond	4.6.6	3.9

4.5.2.2 Group B. - Group B test shall be conducted on filters selected from units which have passed the examination and tests in 4.5.1 and 4.5.2. Sampling shall be in accordance with MIL-STD-105, AQL 2.5 percent defective, inspection level S-3. Tests shall be conducted in accordance with Table IV, and may be performed in any sequence. Failure of any test shall be cause for rejection.

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TABLE IV GROUP "B" TEST SCHEDULE

TEST	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Spectral Transmission (to be conducted at 70° to 80° F)	4.6.2	3.6

4.5.2.3 Group C. - This inspection shall consist of the tests specified in Table V and shall be performed on sample units that have been subjected to and met groups A and B inspection. Sample units shall be selected in accordance with 4.5.2.3.1. Failure of any test shall be cause for rejection.

4.5.2.3.1 Sampling for Inspection. - One sample filter shall be selected at random for all inspections in Table V from each lot of 100 filters produced or every sixty days, whichever occurs first. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

TABLE V GROUP"C"TESTS

TESTS	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Environmental	4.6.3	3.10
Interference Coating Radiation Exposure	4.6.4	3.7
Visual Security	4.6.5.1	3.8

4.5.2.3.2 Noncompliance. - The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has

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been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group C reinspection results have shown, that the corrective action was effective.

4.5.3 Reinspection of Conforming Group B and Group C Sample Units. Unless otherwise specified, sample units which have been subjected to and passed both group B and group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Test Procedures. - Tests required to determine compliance with this specification shall be conducted in accordance with the test procedures specified herein. All meters used during tests shall have been calibrated within 30 days prior to the start of testing, and thereafter as required, and at intervals not to exceed 6 months. Spectral transmission measurements shall be made using a spectrophotometer calibrated to read wavelength from 400 millimicrons to 1,100 millimicrons. The spectrophotometer shall be capable of measuring transmission from 100 percent to 0.01 percent from 400 millimicrons to 1,100 millimicrons, and shall be calibrated to read the transmission within plus or minus 3 percent for any wavelength. Ambient temperatures and operating temperatures during environmental testing shall be recorded using recording instruments having not less than 1.0 percent accuracy. All tests except the interference coating bond shall be conducted on the complete filter.

4.6.1 Pin Holing. - Each individual filter segment shall be examined for pin holes in the interference coating. The filter segment shall be supported in a test fixture so that no more than 1/8 inch of the edge of the flat surface shall be concealed by the supporting frame. The coated surface shall be exposed to the test personnel during this test. An incandescent lamp of not less than 250 watts shall be located 36 to 40 inches behind the filter segment. The test fixture shall not permit passage of visible light except through the filter segment. The test observer shall be in the darkened test room not less than 10 minutes prior to conducting the pin hole test to become dark adapted, and shall be located not more than 40 feet distant from the filter segment being tested. With the incandescent lamp "ON" the filter segment shall be examined

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for leakage of visible light through pinholes, cracks or other defects in the interference coating. Any defect in the coating which permits passage of visible light shall be masked with opaque ceramic paint. When all visible light emission has been masked the filter segment shall be removed to a well lighted room and the total area masked by the opaque ceramic paint shall be measured. The percentage of the filter segment marked shall be calculated by the following:

$$\text{Percent of Filter Area} = \frac{A_1}{A_2}$$

Where: A_1 = masked area and A_2 = segment area

Any evidence of visible light leakage after applying the opaque ceramic paint or masking of more than 0.1% of the total filter segment area shall constitute failure of this test.

4.6.2 Spectral Transmission. - The spectral transmission of the infrared filter segments shall be measured at not less than four points and at 90 degrees from the surface of the filter. Spectral transmission measurements shall be made with the outside surface of the filter at 70° to 80° F and again at 290° F to 300° F or equilibrium temperature attained in an approval searchlight sub-assembly. The spectral transmission curves for each point measured on the filter shall be plotted on one sheet of 5-cycle semi-log paper. Any apparent shift of spectral transmission of usable zone of filter. Failure to meet the spectral transmission requirements of figure 1 or any apparent shift in excess of the limits specified in figure 1 shall constitute failure of this test.

4.6.3 Environmental. - The filter shall be stored in an ambient temperature of 165° F with maximum relative humidity, for a period of 4 hours and then examined for damage (see 6.3). The filter shall then be stored at an ambient temperature of minus 80° F for a period of 4 hours, after which the ambient temperature shall be raised to minus 65° F and maintained for 4 hours. The temperature shall then be raised within a period of 10 minutes to a temperature of 300° F and maintained for 45 minutes. The filter shall then be cooled to a temperature of 85° F $\pm$ 15° F and examined for damage. Any evidence of damage shall constitute failure of this test.

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4.6.4 Interference Coating Radiation Exposure. - The filter shall be exposed to the radiation from a 20 KW xenon lamp for 20 hours maintaining the outside filter temperature of 295° F ± 5° F. After 20 hours of exposure to the lamp radiation, the spectral transmission of the usable zone of the filter shall be tested in accordance with 4.6.2 with the filter at a temperature of plus 295° F ± 5° F. The filter shall then be cooled to a temperature of plus 75° F ± 5° F and the spectral transmission of the filter shall be tested in accordance with 4.6.2. Failure of the filter to meet the requirements of 3.6 shall constitute failure of this test.

4.6.5 Visual Security. - The filter shall be mounted in a mechanism simulating the xenon short-arc searchlight AN/TVS-3. The mechanism shall be operated on a suitable outdoor range at night with ambient light conditions not to exceed 1×10^{-4} foot candles at the 20 KW xenon power level for 30 minutes prior to determining the visual security distance. Three observers shall be dark adapted for at least 30 minutes while the filter is temperature stabilizing. The observers shall have 20/20 vision or corrected to 20/20 vision and shall have no defects of vision which would affect the validity of the test. After the dark-adaptation period, each observer shall look at the filter in the center of the beam from a point 1200 meters from the searchlight. Each observer shall then decrease the distance between himself and the light until the light can be detected by visual observation. The point where each observer can see the light shall be marked and the distance measured from this point to the filter. An average of the three distances greater than 1000 meters or less than 800 meters shall constitute failure of this test. The filter shall then be removed from the mechanism and placed on a test fixture over a variable intensity light source, located at the focal point of a reflector conforming to MIL-R-52341A. The same three observers shall be placed 20 feet from the surface of the filter and the room darkened for not less than 30 minutes to enable the observer to become dark adapted The light source shall be turned on and the light intensity increased until 2 of the 3 observers can detect the light through the filter. The light level shall be measured and recorded.

4.6.5.1 Group C Tests. - The filter shall be installed on the test fixture used in 4.6.5 and the light level raised to the level recorded in 4.6.5. Three observers with normal or corrected vision and having no defects of vision which would affect the validity of this test shall be located not less than 30 feet

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from the surface of the filter kit and the room shall be darkened for 30 minutes. The observers shall walk slowly toward the filter until the light is detected. The point where each observer can see the light shall be marked and the distance measured from this point to the filter. An average distance greater than 22 feet or less than 18 feet shall constitute failure of this test.

4.6.6 Interference Coating Bond. - A strip of clear 3/4 inch cellophane tape approximately 10 inches long shall be applied to the interference coating and allowed to set for 5 minutes. At the end of 5 minutes, a 290-gram weight shall be attached to the free end of the tape and the weight allowed to hang free of the interference coating surface at an angle of 90 degrees from the surface for a period of 5 minutes. Then the tape shall be pulled sharply off the coating at approximately a 90 degree angle. Any evidence of interference coating on the tape shall constitute failure of this test.

4.7 Inspection comparison. - The Government may select parts, assemblies, or complete filters at any time during the contract period and subject the parts, assemblies or complete filters to the examination specified in Table I and the applicable test specified in Table II to reveal any manufacturing deficiencies and for conformance to prescribed requirements. The tests shall be performed by the Government at a site selected by the Government.

4.7.1 Failure. - Failure of any part, assembly or complete filter when examined and tested as specified, shall be cause for refusal by the Government to continue acceptance of parts, assemblies or complete filters until objective evidence has been provided by the contractor that the deficiency has been corrected.

4.8 Inspection of Preparation for Delivery. - Preservation, packaging, packing and marking as specified in 5.1 shall be inspected as specified in MIL-E-17555. Failure to pass any portion of the inspection shall constitute failure of the test.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, Packaging, Packing and Marking. - Filters shall be portected, packaged, packed and marked in accordance with MIL-E-17555 for level A or C packaging as specified (see 6.2) and level A, B, or C packing as specified (see 6.2).

6. NOTES

6.1 Intended Use. - The interference infrared filter is intended for use in the infrared filter kit for the 20 KW xenon searchlight, AN/TVS-3.

6.2 Ordering Data. - Procurement documents should specify the following:

a. Title and date of this specification.

b. Level of preservation and packaging and level of packing required (see 5.1).

6.3 Damage. - Damage is defined as chipping, breaking, pitting and cracking or peeling.

CUSTODIAN:

ARMY - EL

PREPARING ACTIVITY:

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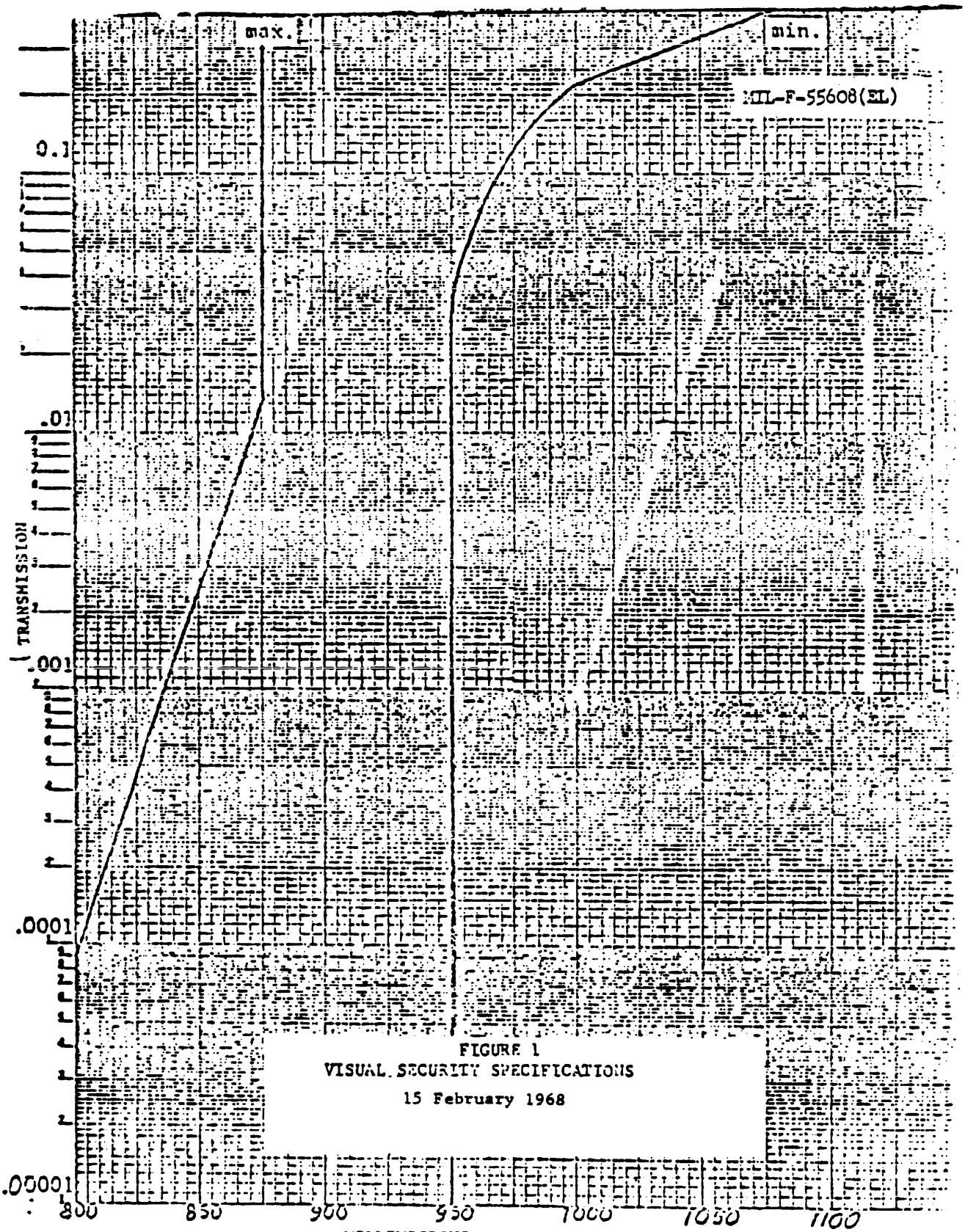


FIGURE 1
VISUAL SECURITY SPECIFICATIONS
15 February 1968

SPECTRAL TRANSMISSION OF USABLE ZONE OF FILTER

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-F-55608(RL) FILTER, INFRARED, INTERFERENCE TYPE, FOR AN/TVS-3

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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